

Article

Efficient Design of Reversible Adder and Multiplier Using Peres Gates

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Abstract: This paper details the approach to the efficient design and optimization of a reversible adder and multiplier utilizing Peres gates, which is a three-input, three-output gate. Peres gates are recognized for their universality and energy-efficient properties and present an intriguing option for constructing reversible circuits. Reversible logic is characterized by its ability to uniquely determine input states from output states. The design methodology involves a cascading arrangement of Peres gates, each performing a reversible XOR operation on corresponding bits of the input numbers and the carry-out from the preceding stage. This paper presents a detailed schematic representation, simulation results, and analysis of the proposed designs of different adders and multipliers, showcasing their potential for reversible applications. The integration of Peres gates in the adder and multiplier design signifies a step forward in the exploration of reversible logic circuits and their applications in contemporary computing paradigms. The overall hardware reduction is the main achievement of this research in terms of quantum cost.

Keywords: carry-save adder; carry skip adder; look ahead carry adder; multiplier; Peres gates; ripple carry adder; reversible logic



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1. Introduction

The ever-evolving landscape of digital computing demands innovative solutions to address the challenges posed by emerging technologies. Reversible logic has gained prominence for its unique ability to perform computations in a manner that enables the retrieval of input states from output states [1]. The potential uses of reversible logic include low-power computing, where large-scale data centers and energy-efficient computing reduce overall power consumption, leading to greener and more sustainable data processing [2,3]. In quantum computing, error-correcting codes and reversible logic help in creating fault-tolerant quantum circuits [4,5]. Since no information is lost, reversible logic facilitates the complete and verifiable erasure of data, which is critical for privacy and security. Reversible logic also has applications in DNA-based computation [6] and molecular computing [7], useful in areas like drug discovery and bio-molecular sensing. The nano-electronics devices operate at less power, having nanotechnology applications [8]. Reversible logic circuits are being studied as an alternative to classical CMOS circuits, potentially leading to more compact and efficient chip design. Reversible logic plays an important role in creating more energy-efficient artificial intelligence accelerators and neural networks [9]. The adders, multipliers, and other arithmetic units constructed from reversible logic lead to improvements in digital signal processing and digital image processing [10]. Reversible logic has the potential to revolutionize the computing world by enabling more energy-efficient, sustainable, and scalable systems across a wide range of industries.

Reversible logic is a field of study on developing circuits where the input is uniquely recovered from the outputs, which means there is no loss of information during computation [11]. Some of the state-of-the-art technologies and approaches in reversible logic

include quantum technologies, adiabatic logic, CMOS-based reversible circuits, emerging reversible logic gates, fault-tolerant reversible computing, reversible logic synthesis, and neuro-morphic and beyond CMOS computing.

The quantum gates are reversible by nature. The surface code uses reversible logic to minimize errors during quantum computations, as qubits are prone to errors from external noise [12]. Adiabatic logic circuits are increasingly used in applications requiring minimal power consumption, such as battery-operated devices, sensor networks, and wearables. Techniques like dual rail logic and pass transistor logic are employed to build reversible circuits that are compatible with existing CMOS technology. Reversible logic is being applied to nano-electronics to develop extremely small, low-power, high-performance computational devices. Technologies like quantum dot cellular automata (QCA) and single electron transistors (SETs) are being used to explore reversible logic at the nanoscale [13,14]. Reversible circuits are also explored for fault tolerance in computing systems. These designs are less prone to errors compared to conventional irreversible logic circuits, which makes them attractive for mission-critical applications like space computing and medical devices. Research is being conducted into photonic circuits that use light instead of electricity to implement reversible logic. Optical reversible logic gates such as Mach–Zehnder interferometers perform computation with lower energy dissipation [15]. Reversible logic is also applied in quantum photonic computing, where photons are used as information carriers to implement efficient and fast reversible operations. Advanced software tools are being developed to automatically synthesize reversible circuits from high-level descriptions [16]. These tools focus on minimizing gate count and quantum cost, which are critical metrics in reversible computing. Ongoing research is directed at optimizing reversible circuits in terms of quantum cost, gate count, etc. Automated algorithms for synthesizing and optimizing reversible circuits are becoming more sophisticated. Technologies like memristors and spintronics are being explored for implementing energy-efficient reversible logic [17]. These are the key components in neuromorphic computing and brain-inspired architectures that aim to mimic the efficiency of biological neural networks [18].

Quantum cost in reversible logic refers to the number of quantum gates required to implement a reversible logic function using quantum technology. This metric helps to assess the efficiency of implementing logical operations in quantum computing since reversible logic circuits can be transformed into quantum circuits. Gate count is related to counting the specific quantum gates needed, like CNOT, Toffoli, etc., to achieve the desired function. Complexity is used to evaluate how the number of gates scales with the size of the input or the complexity of the logic function.

Minimizing quantum cost is crucial for optimizing quantum algorithms, as it can impact both the speed of computation and the resources required, like qubit usage and error rates.

The ongoing efforts focus on minimizing energy dissipation and achieving close to zero power operation. Researchers are working on scalable designs that handle large-scale reversible computations without an exponential increase in complexity. In quantum computing, the use of reversible logic is critical for maintaining quantum coherence and maximizing computational efficiency. These cutting-edge technologies and approaches help to advance the field of reversible logic, pushing the boundaries of energy-efficient and quantum-based computing systems.

The ripple carry adder is a fundamental digital circuit used for binary addition. This paper explores the utilization of Peres gates in the design of a ripple carry adder and other adders aiming to leverage the reversible properties of Peres gates for potential applications in quantum computing where the preservation of information is a critical consideration [19]. Multipliers are fundamental components in digital circuits responsible for executing arithmetic operations, with their efficiency playing a pivotal role in overall system performance. Integrating Peres gates into multiplier design offers a promising avenue for achieving both computational precision and energy efficiency.

Peres gates, characterized by their reversible nature, present an intriguing opportunity to create circuits that can be seamlessly integrated into the quantum computing paradigm. This paper endeavors to investigate the design and implementation of an RCA using Peres gates and the specific design methodology employed in constructing a 4×4 reversible multiplier using Peres gates, aiming to harness the benefits of reversibility for enhanced computational capability.

The structure of this paper is organized as follows: Section 2 provides an overview of reversible logic and the unique characteristics of Peres gates. Section 3 details the design methodology, outlining the steps involved in constructing the RCA, CSA, and multiplier using Peres gates. Section 4 details the design methodology, outlining the steps involved in constructing the multipliers with RCA and CSA using Peres gates. Section 5 presents a schematic representation of the designs illustrating the interconnection of Peres gates and input/output bits. Simulation results and analysis are discussed in Section 6, highlighting the performance and potential applications of the proposed design. The paper concludes in Section 7 with a summary of findings and suggestions for future research directions.

This exploration into the integration of Peres gates in a fundamental arithmetic circuit holds promise for advancing the field of reversible and quantum computing, contributing to the ongoing efforts to redefine the capabilities of digital systems in the face of evolving computational paradigms.

2. Reversible Logic and Peres Gates

Reversible logic stands out in the realm of digital circuit design due to its distinctive property of information preservation, allowing for computations where original input states can be uniquely reconstructed from the corresponding output states.

This reversibility characteristic becomes particularly relevant in quantum computing, where the conservation of information is crucial for leveraging quantum superposition and entanglement. Peres gates are particularly valuable in reversible logic design because they facilitate the creation of circuits that adhere to the principles of information conservation. Peres gate [20] with three inputs and three outputs are shown in Figures 1–3. Both Figures 2 and 3 represent the Peres gate, but the difference between these two representations is that the Peres gate in Figure 2 has a gate count of 1, and the number of the two-qubit model is 4, whereas in Figure 3, where the Peres gate is constructed using one-three input Toffoli gate followed by one-two input Toffoli gate, the gate count is 2 and the number of the two-qubit model is 6 [21]. So, instead of one Peres gate, if we use two Toffoli gates to construct all the circuits discussed in this paper, then we need twice the number of Toffoli gates as compared to the Peres gate, and also, the quantum cost will further increase by 50% for the same. Similarly, comparing the quantum cost of the Fredkin gate with the Peres gate, there is an increase of 25% in the quantum cost of the Fredkin gate as compared to Peres gates, which further increases the quantum cost of the applications of adders and multipliers where we are replacing the Peres gates with the Fredkin gates [20]. All the circuits mentioned in this paper are constructed using the Peres gate shown in Figure 2.

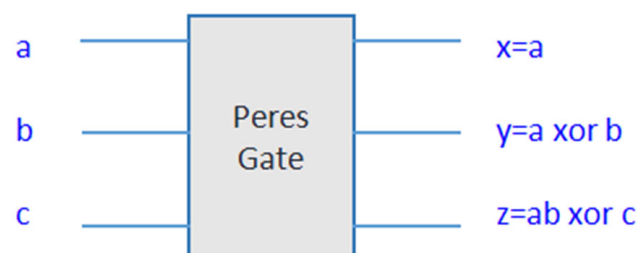


Figure 1. Peres gate.

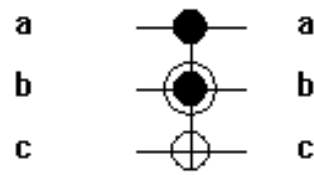


Figure 2. Peres gate quantum circuit.

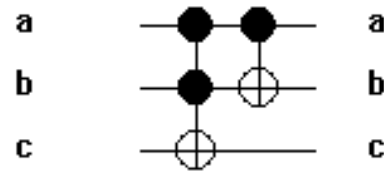


Figure 3. Peres gate quantum circuit using two Toffoli gates.

The truth table in Table 1 describes the relation between inputs and outputs. Output *a* is the same as input *a*; output *b* can be obtained by logically exoring the inputs *a* and *b*, whereas output *c* can be obtained as logical exoring of input *c* with logical ending of *a* and *b*, i.e., ($ab \text{ xor } c$).

Table 1. Truth table for Peres gate.

Input Variables			Output Variables		
<i>a</i>	<i>b</i>	<i>c</i>	<i>a</i>	<i>b</i>	<i>c</i>
0	0	0	0	0	0
0	0	1	0	0	1
0	1	0	0	1	0
0	1	1	0	1	1
1	0	0	1	1	0
1	0	1	1	1	1
1	1	0	1	0	1
1	1	1	1	0	0

In the context of the ripple carry adder (RCA) design, Peres gates are employed to perform reversible XOR operations, which are fundamental to binary addition. The ability to selectively apply NOT operations based on control bits makes Peres gates a versatile building block for constructing circuits with minimal energy dissipation, a critical consideration in quantum and reversible computing.

The reversible nature of Peres gates ensures that the input states can be accurately reconstructed from the output states, aligning with the requirements of quantum computing algorithms. The controlled-NOT operation provided by Peres gates enables the creation of a coherent and efficient RCA design, contributing to the exploration of reversible logic in practical applications. This section sets the stage for the subsequent discussion on the design methodology where the unique characteristics of Peres gates are leveraged to construct reversible ripple carry adder, carry-save adder, and reversible multipliers using these two adders. The use of reversible logic, particularly through Peres gates, presents an exciting avenue for the advancement of computing paradigms, including quantum and emerging reversible computing technologies.

3. Design Methodology

This section details the design methods used for the development of the Peres gate and different adder and multiplier circuits, including the Vedic multiplier using reversible

logic. The design of the ripple carry adder, carry-save adder, look ahead carry adder, carry skip adder, and carry bypass adder [22–24] using Peres gates follows a systematic and staged approach leveraging the reversible nature of Peres gates to achieve an efficient and coherent circuit. The methodology involves the incorporation of Peres gates in a cascading fashion to perform reversible XOR operations on corresponding bits of the input numbers and carry-outs from the preceding stages.

Reversible logic gates enable the creation of a coherent and efficient RCA design, contributing to the exploration of reversible logic in practical applications. This section sets the stage for the subsequent discussion on the design methodology where the unique characteristics of Peres gates are leveraged to construct a reversible ripple carry adder, carry-save adder, and reversible multipliers using these two adders. The use of reversible logic, particularly through Peres gates, presents an exciting avenue for the advancement of computing paradigms, including quantum and emerging reversible computing technologies.

The design of the ripple carry adder and carry-save adder using Peres gates follows a stage-wise approach:

- **First Bit (LSB):** A Peres gate is employed to perform a reversible XOR operation on the least significant bits of the numbers to be added and the input carry. This stage produces the sum bit and a carry-out for the next stage.
- **Intermediate Bits (1 to $2*n-1$):** Peres gates are used in a cascading fashion where each gate performs a reversible XOR operation on the corresponding bits of the input numbers and the carry-out from the previous stage. This process is repeated for the intermediate bits.
- **Final Bit (MSB):** The most significant bits of the input numbers and the carry-out from the previous stage are fed into a Peres gate to produce the final sum bit and carry-out.

The output of the RCA is the concatenation of the sum bits ($S[2*n-1:0]$). The final carry-out ($Cout_n$) serves as an overflow indicator, providing information about whether the addition operation resulted in a carry beyond the designated bit size.

Peres gates, which belong to the class of three input quantum gates, have shown promise in implementing various logical functions, and we focused on utilizing Peres gates to design 4×4 reversible multipliers [25–32]. Peres gates introduced by Asher Peres in 1984 have been recognized for their universality, meaning they can be used to construct any reversible circuit [32].

This characteristic makes Peres gates an attractive choice for designing complex reversible circuits such as multipliers. The 2×2 and 4×4 reversible multipliers are constructed using a combination of Toffoli gates and Peres gates.

The simplest form of multiplier circuit formed as a 1-bit multiplier is with a single Peres gate with multiplicand as a_0 and multiplier as b_0 , and the third input given to the Peres gate will be the product result as the output given by c_0 , as shown in Figure 4. The product obtained at the output is 0 when multiplicand a_0 is 0 or multiplier b_0 is 0, and the maximum value of the product is 1 when both a_0 and b_0 are at logic 1. In Figure 5, variable a is considered to be a multiplicand, and variable b is a multiplier. The product of a two-bit number is 0 when the multiplier is 0, and multiplicand has any value between 0 and 3, whereas the maximum value of the product will be 6 when both multiplicand and multipliers have decimal value 3 or “11” in binary format. A four-level decomposition of this two-bit multiplier is shown in Figure 6.

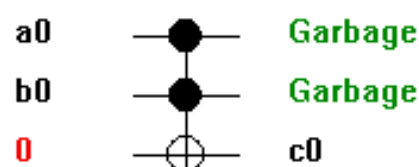


Figure 4. One bit multiplier.

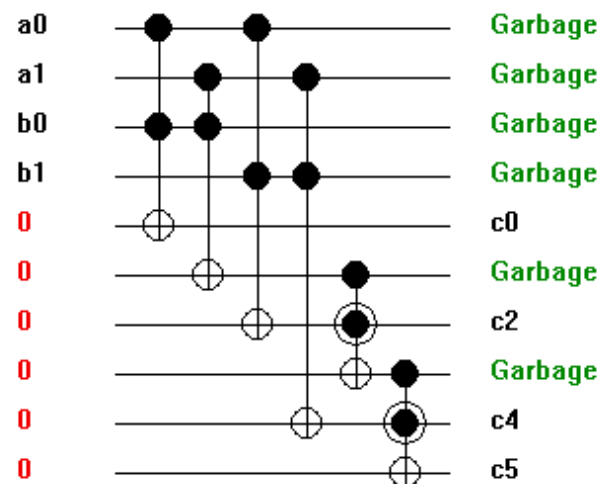


Figure 5. Two-bit multiplier quantum circuit using Peres gates.

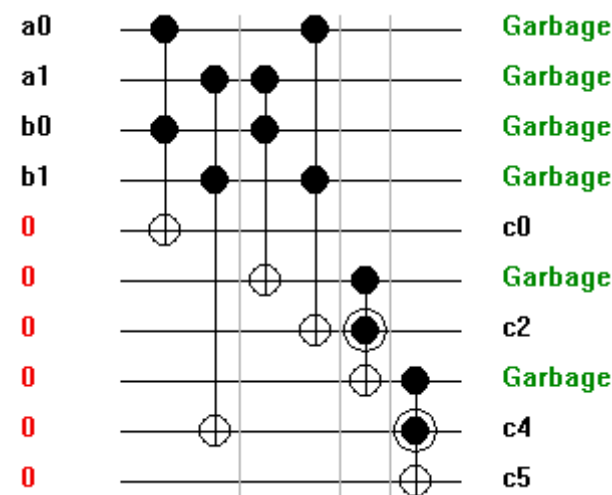


Figure 6. Four-level decomposition of this two-bit multiplier.

4. Schematic Representation

A schematic diagram visually represents the connections between Peres gates, input bits (A, B), carry-ins, carry-outs, and the sum bits in the cascading arrangement of information preservation.

This design methodology forms the foundation for the subsequent sections, where a detailed schematic representation, simulation results, and analysis will be presented to validate the functionality and performance of the proposed adders using Peres gates. Schematic diagrams illustrate the connections between Peres gates, input bits, and carry-outs at each stage, as depicted in Figures 7 and 8, representing the 2-bit Vedic multiplier and its 4-level decomposition, respectively. The reversible nature of Peres gates allows for a coherent design that ensures the reversibility of the overall circuit. Peres gates highlight the systematic application of reversible XOR operations at each stage, ensuring the overall circuit adheres to the principles of reversible logic. Figures 9–13 represent the generalized full adder, 1-bit full adder block schematic, 1-bit full adder using Peres gates, 1-bit adder quantum circuit using Peres gates, generalized block schematic of the n-bit ripple carry adder, and simplified block schematic of the n-bit ripple carry adder, respectively.

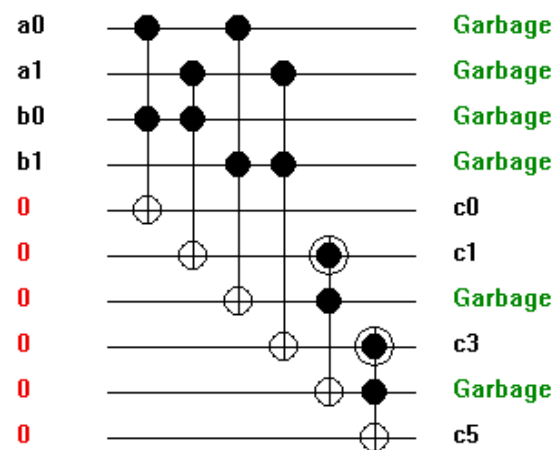


Figure 7. Two-bit Vedic multiplier quantum circuit using Peres gates.

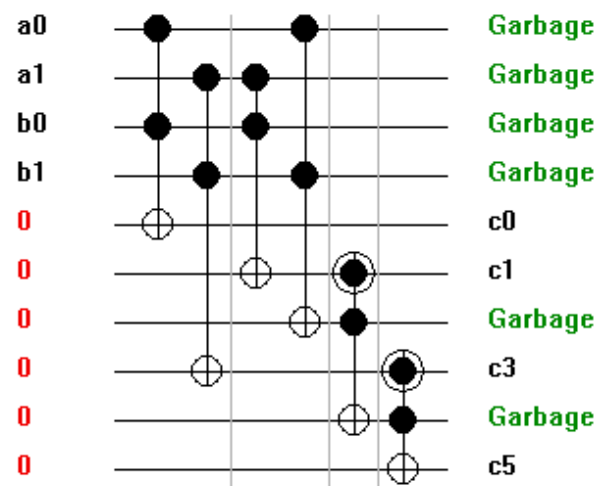


Figure 8. Four-level decomposition of this 2-bit Vedic multiplier.

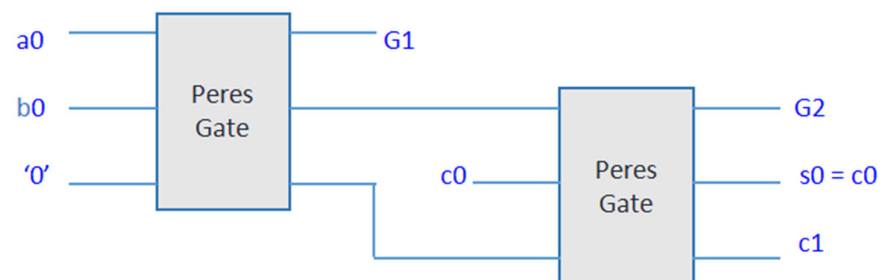


Figure 9. Full Adder using two Peres gates.

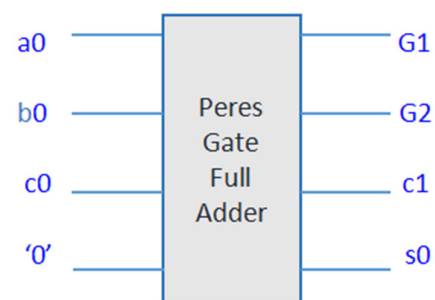


Figure 10. One-bit adder block schematic using Peres gates.

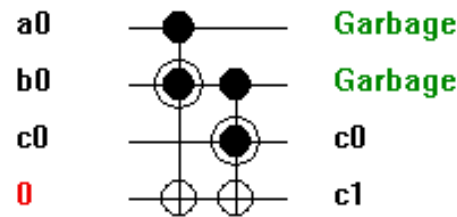


Figure 11. One-bit adder quantum circuit using Peres gates.

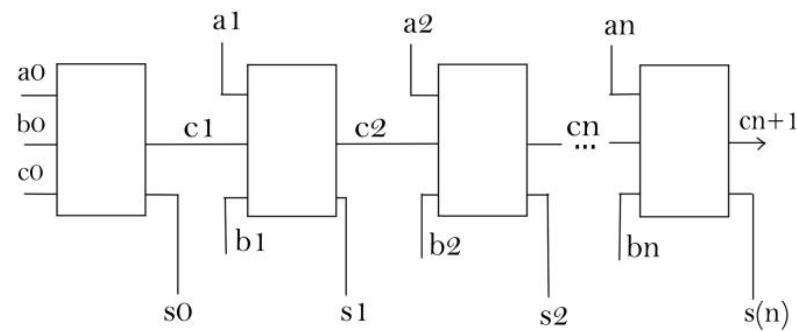


Figure 12. Generalized block schematic representation of an n-bit ripple carry adder.

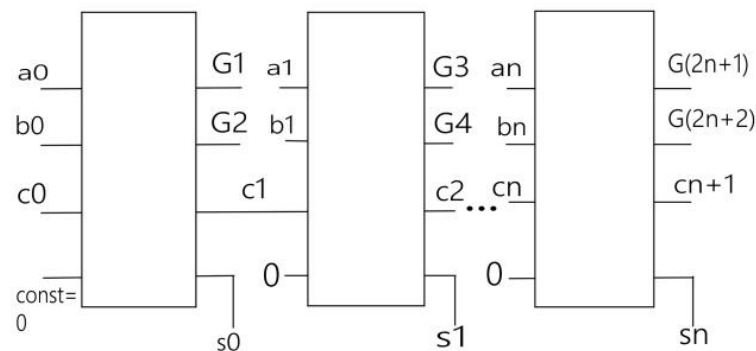


Figure 13. Block schematic representation of an n-bit ripple carry adder using Peres gates.

The use of Peres gates in this design methodology not only ensures reversibility but also opens the door to potential applications in quantum computing due to their inherent characteristics. A schematic diagram illustrates the connections between Peres gates, input bits, and carry-outs at each stage, as depicted in Figures 6 and 7. The reversible nature of Peres gates allows for a coherent design that ensures the reversibility of the overall circuit.

The schematic representation visually captures the interconnection of Peres gates, input bits (A, B), carry-ins (C_{in}), and carry-outs (C_{out}) at each stage of the adders. The design utilizes Peres gates for reversible operations, ensuring information conservation throughout the addition process. The connections illustrate the systematic application of Peres gates, where each gate performs a reversible XOR operation on the corresponding bits of the input numbers and the carry-out from the previous stage.

The final sum bits ($S[n:0]$) and carry out (C_{outn}) collectively form the output of the full adders. The cascading arrangement of Peres gates signifies the stepwise progression through the bits of the input numbers, ensuring a coherent and reversible addition process. The external control lines represent the controlled-NOT operation applied by the Peres gates based on input bits. The proposed ripple carry adder (RCA) design utilizing Peres gates has been subjected to rigorous simulation to evaluate its functionality and performance for different configurations as 4-bit, 8-bit, and 16-bit addition and subtraction are shown in Figures 14–18. The analysis encompasses key aspects such as speed, quantum cost, and adhering to the principles of reversible logic. Figures 14–18 represent the 4-bit subtractors

with outputs difference and borrows, 4-bit adder cum subtractor, 4-bit adder, 8-bit adder, and 16-bit adders, respectively, using Peres gates. The level-wise decomposition of the 4-bit multiplier is shown in Figure 19.

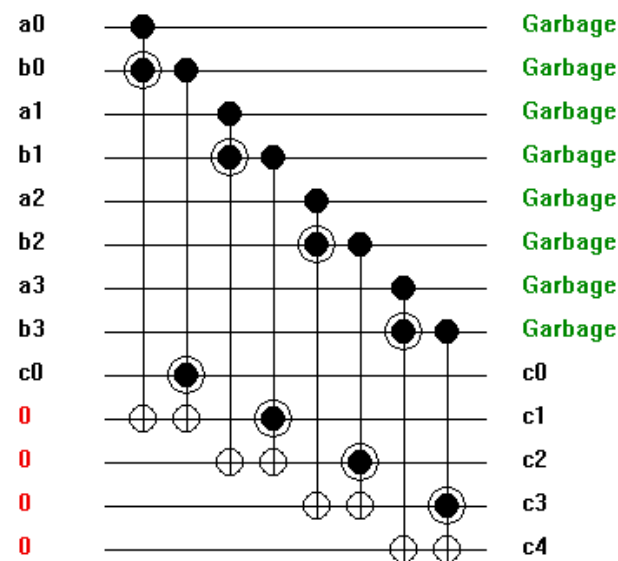


Figure 14. Four-bit subtractor using Peres gates.

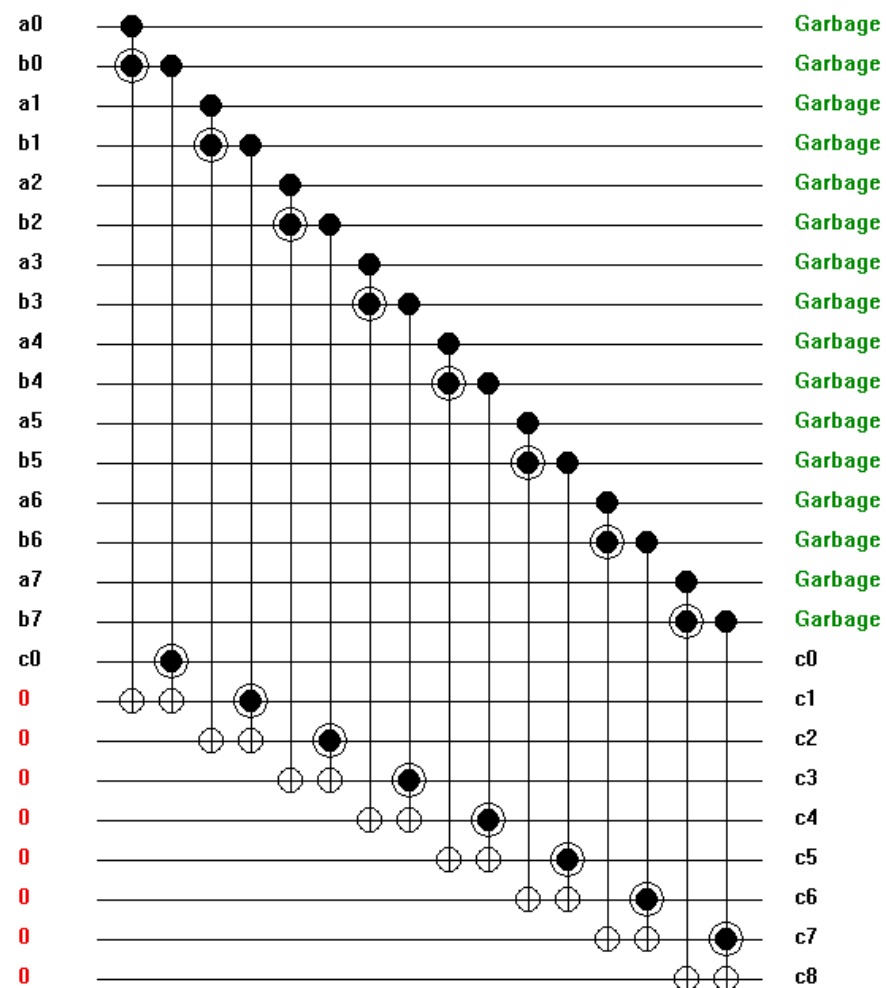


Figure 15. Four-bit adder cum subtractor using Peres Gates.

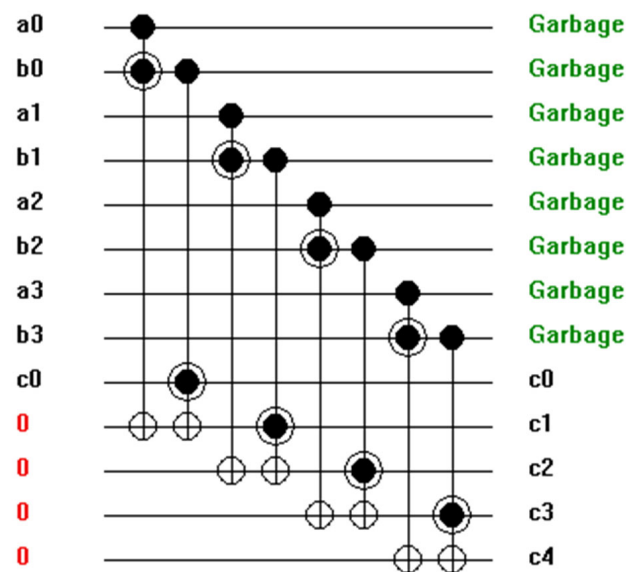


Figure 16. Four-bit adder using Peres gates.

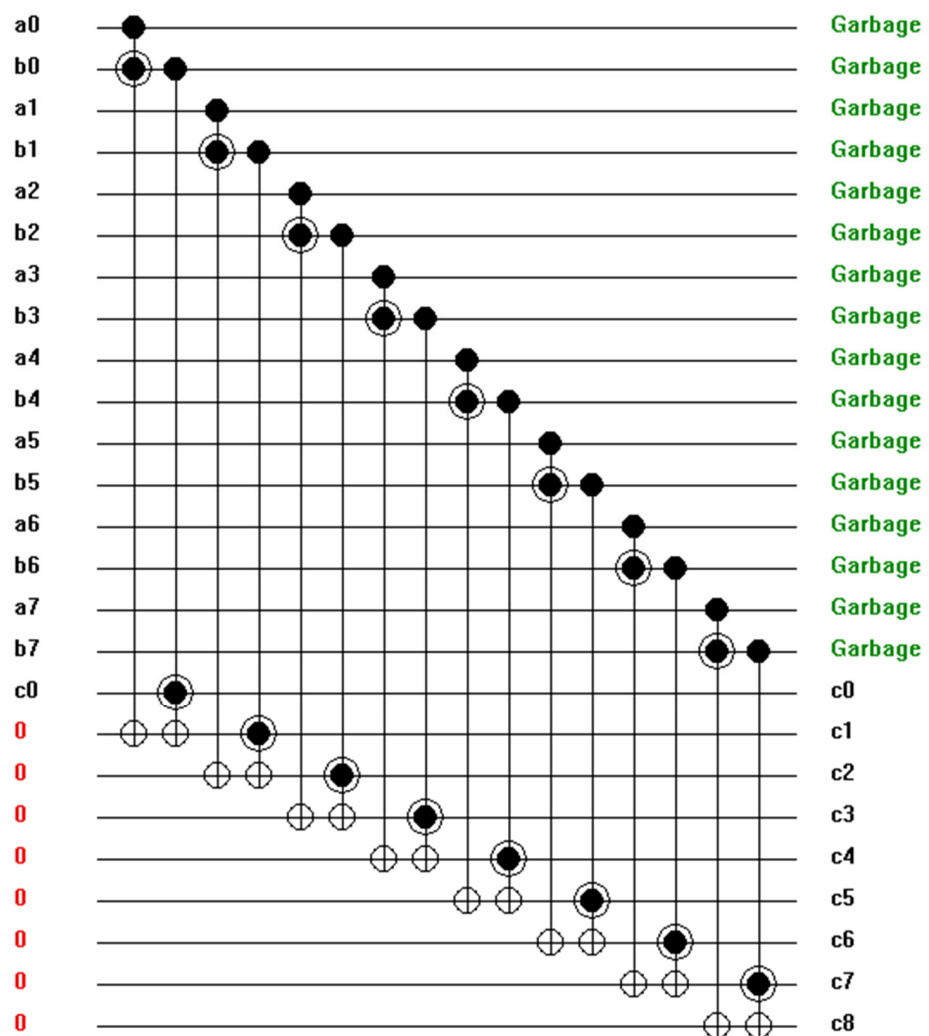


Figure 17. Eight-bit adder using Peres gates.

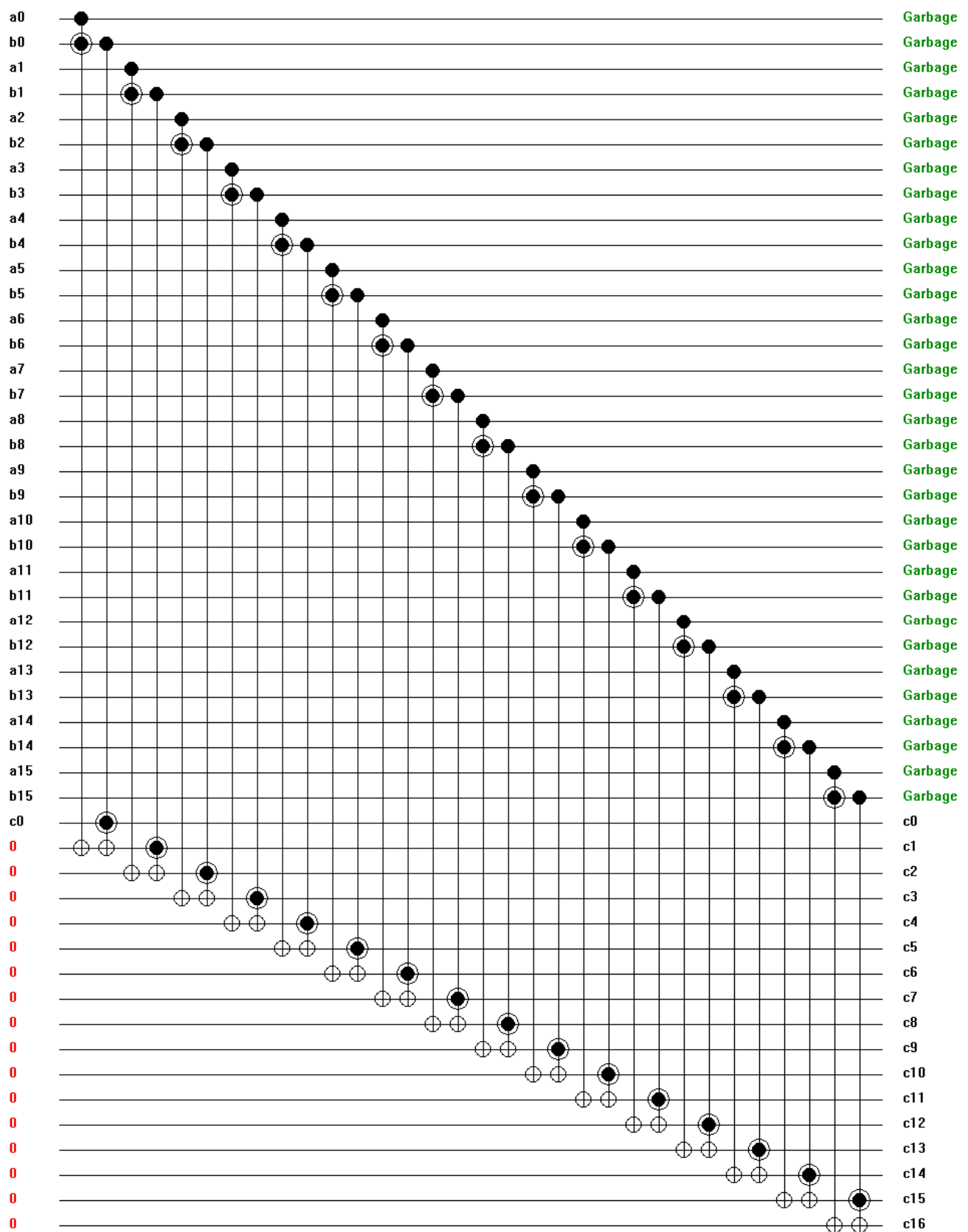


Figure 18. Sixteen-bit adder using Peres Gates.

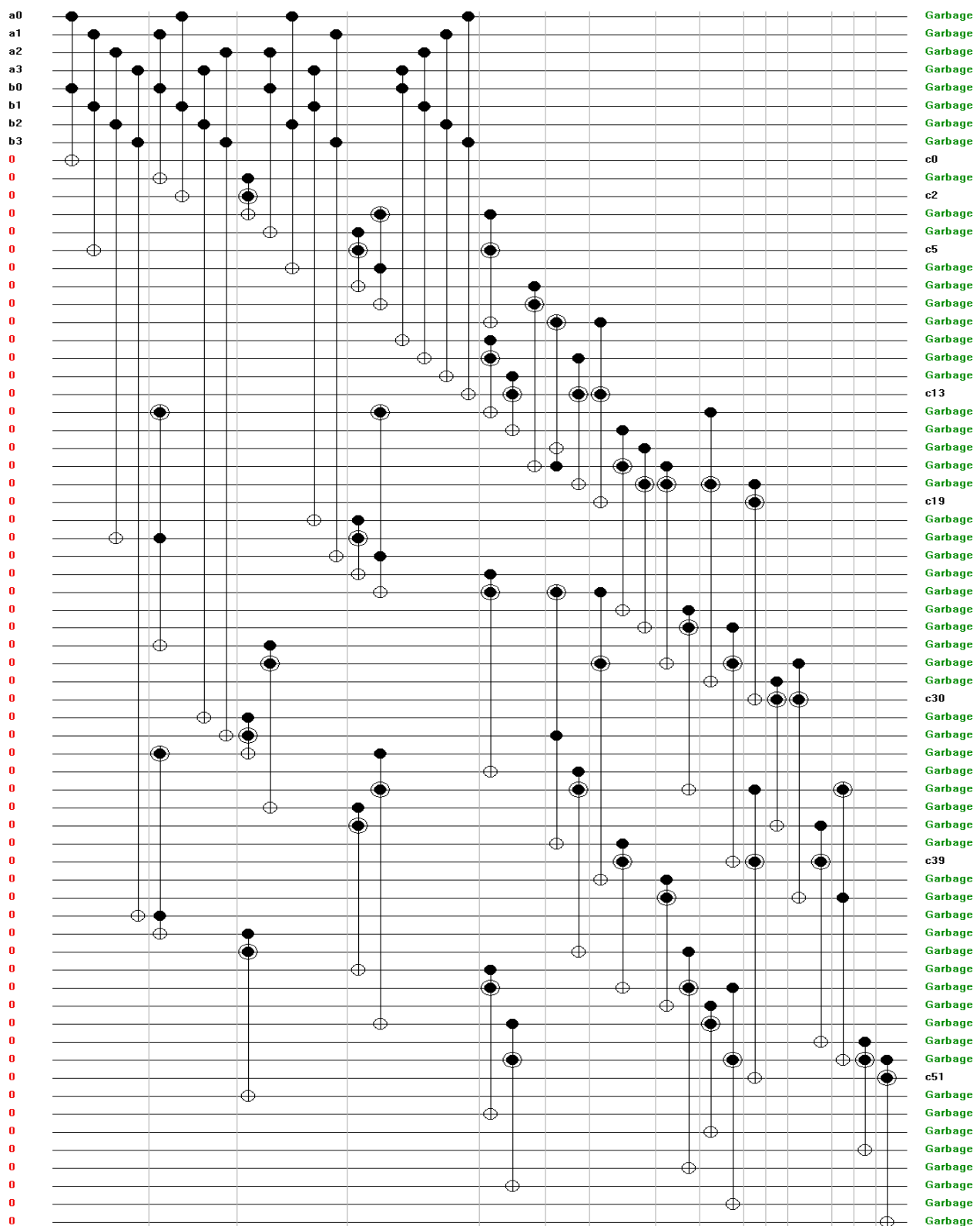


Figure 19. Level-wise decomposition of multiplier configurations using Toffoli and Peres gates.

This schematic representation provides a clear visualization of the reversible logic design, showcasing the utilization of Peres gates to achieve information-preserving binary addition. The subsequent sections will delve into the simulation results and analysis, further validating the functionality and performance of the proposed ripple carry adder using Peres gates.

5. Simulation and Analysis

As a one-step forward in this experimentation, different 4-bit adder configurations were simulated and verified, as shown in Figures 20–23, using ripple carry, carry look ahead, carry save, and carry skip approaches, respectively. Similarly, the designed circuits are simulated to validate their functionality and to analyze the performance in terms of speed and quantum cost. The simulation results demonstrate the successful addition of binary numbers with reversible characteristics.

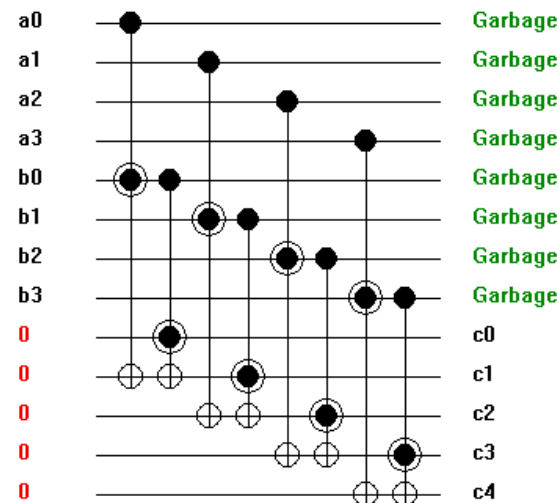


Figure 20. Four-bit ripple carry adder using Peres gates.

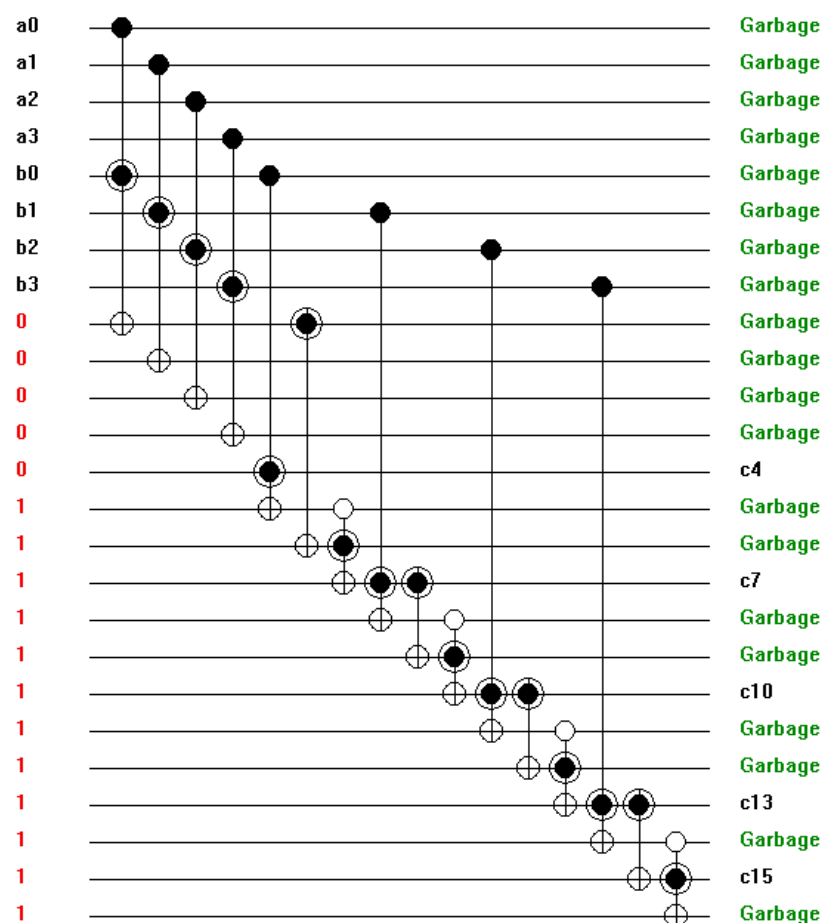


Figure 21. Four-bit carry look-ahead adder using Peres gates.

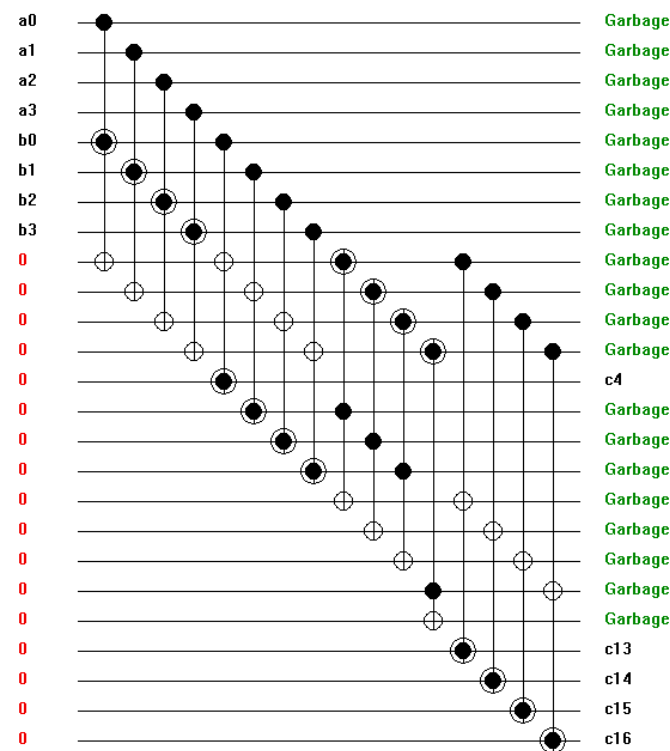


Figure 22. Four-bit carry-save adder using Peres gates.

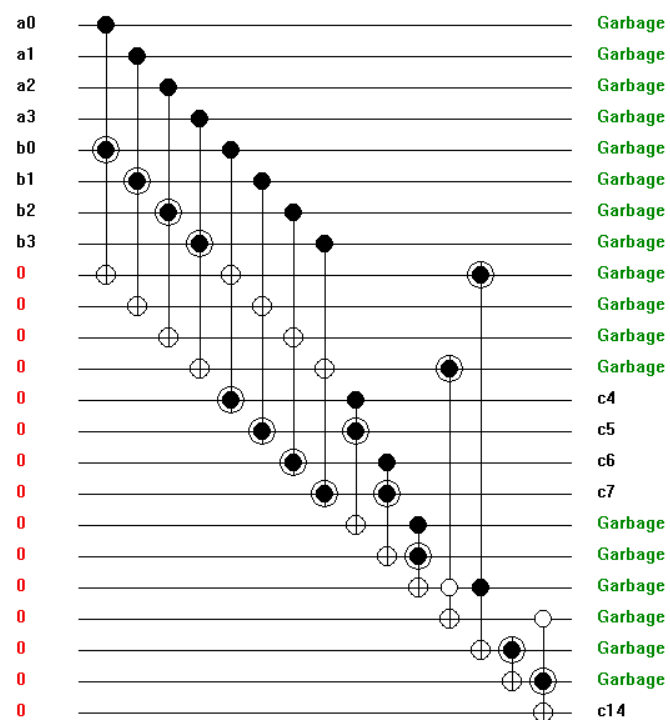


Figure 23. Four-bit carry skip or carry bypass adder using Peres gates.

After simulating all the sub-circuits like reversible AND gate, OR gate, half adder, and full adder, various adders, as mentioned above, were simulated and verified for their functional validity.

The next step is to analyze the 4-bit multiplier circuits using the ripple carry and carry save approach, as shown in Figures 24 and 25, also validated successfully.

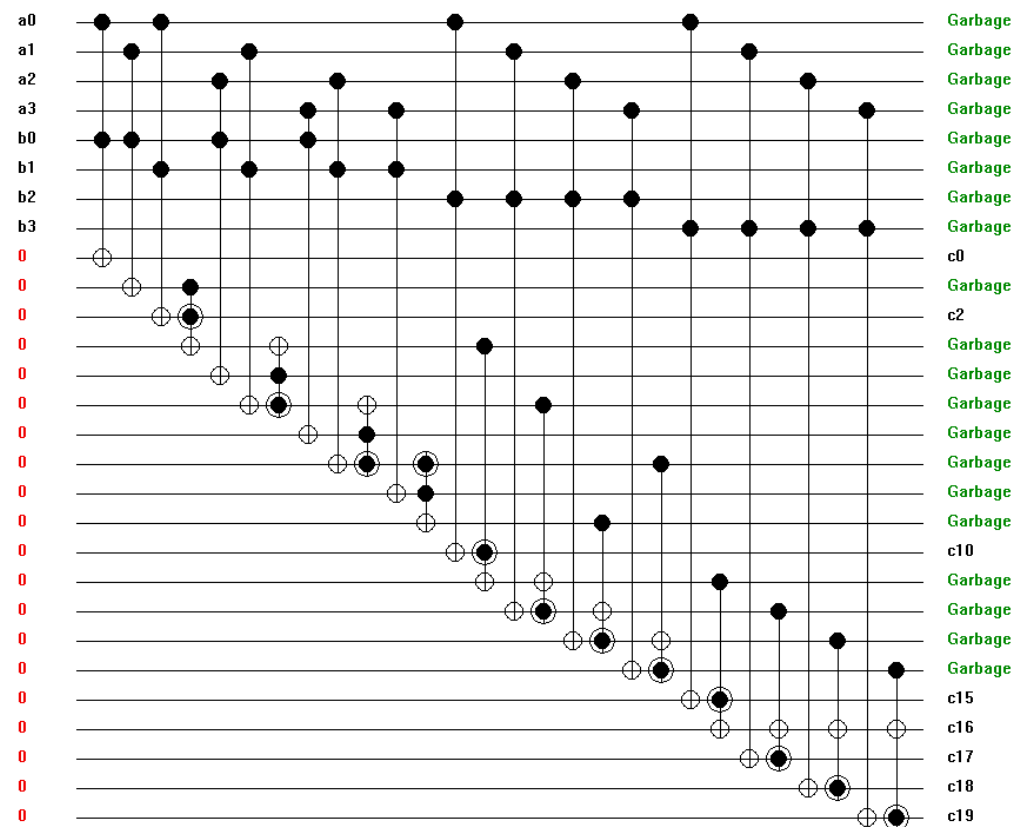


Figure 24. Four-bit multiplier using ripple carry adder.

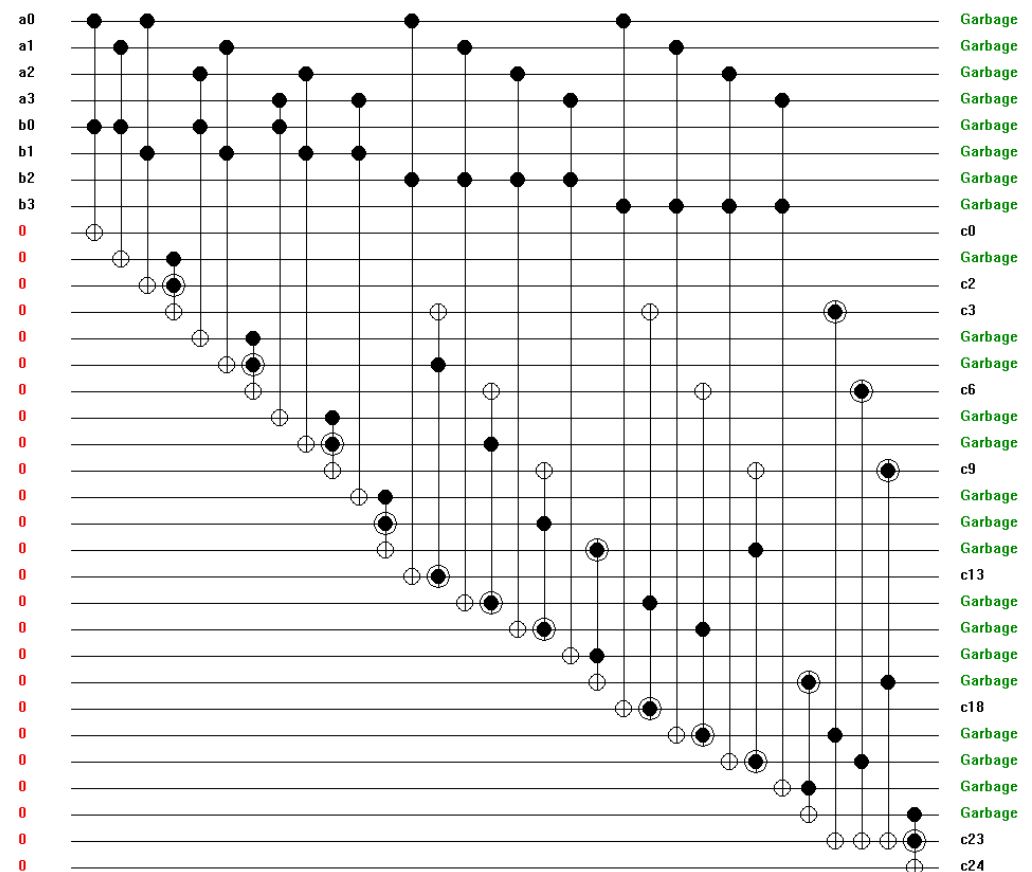


Figure 25. Four-bit multiplier using carry-save adder.

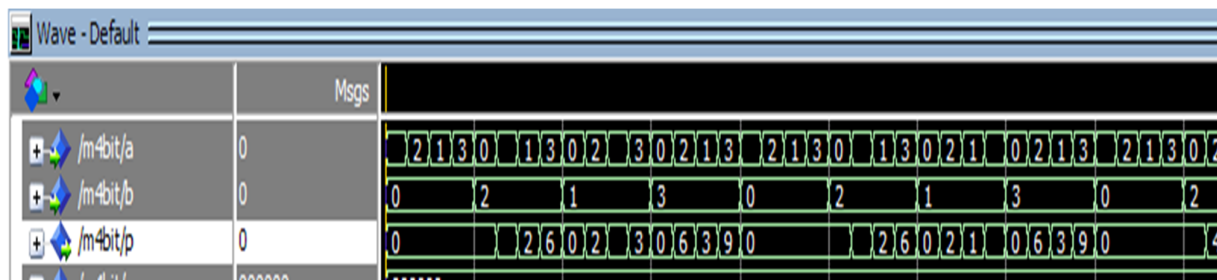


Figure 29. Simulation result for 2-bit multiplier.

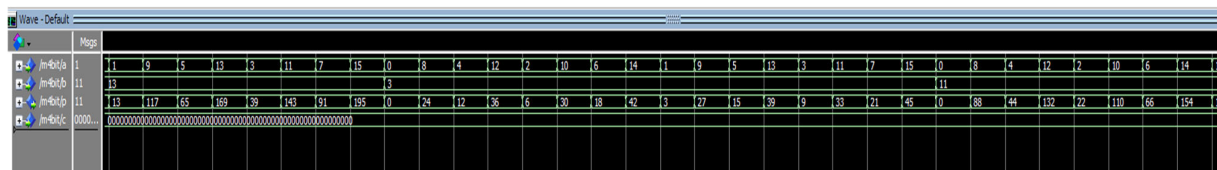


Figure 30. Simulation results for 4-bit multiplier for one input combination.

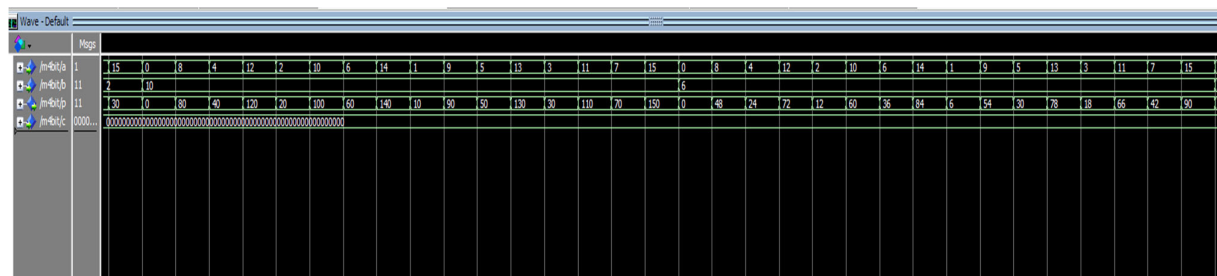


Figure 31. Simulation results for 4-bit multiplier for other input combinations.

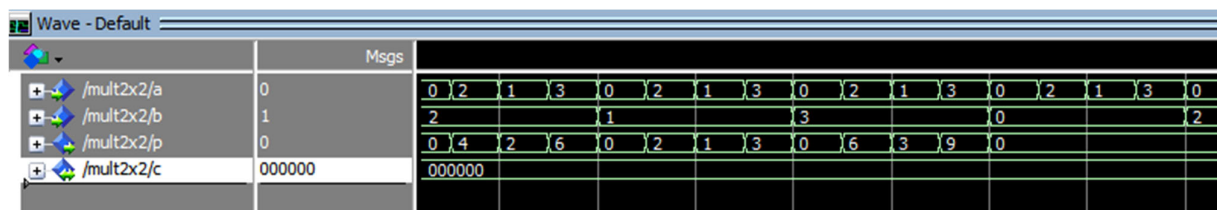


Figure 32. Simulation result for 2-bit Vedic multiplier.

Data Collection

Data collected during the simulations, including verified truth tables, waveforms, timing diagrams, and quantum cost metrics, were stored and analyzed to draw meaningful insights into the behavior of the RCA using Peres gates. The simulation setup serves as a crucial foundation for the subsequent analysis, providing a detailed understanding of how the proposed reversible circuit performs under different conditions and shedding light on its potential advantages in reversible and quantum computing applications.

6. Functionality Verification

The functionality of the ripple carry adder (RCA) design using Peres gates was rigorously verified through simulation, encompassing a range of input scenarios. The simulation results confirm the correct operation of the RCA, producing accurate sum bits ($S[n:0]$) and carry-out ($Cout_n$) for different input combinations. The results confirm the adherence to the reversible logic principles and the preservation of information throughout the addition process. The simulation results demonstrate that the sum bits ($S[n:0]$) produced by the RCA accurately reflect the binary addition of the input numbers (A and B) and the carry-in (Cin).

Each Peres gate strategically placed in the circuit performs a reversible XOR operation, contributing to the generation of the sum bits.

The carry-out (Cout_n) serves as an overflow indicator, revealing whether there is a carry beyond the most significant bit. Simulation results affirm that the circuit correctly calculates Cout_n based on the input numbers and carry-ins, providing crucial information about the completion of the addition operation.

The reversibility of the circuit is a fundamental aspect validated during functionality verification. The Peres gates, by design, enable the reconstruction of input states from output states. Simulation results confirm that the RCA and multiplier design exhibits reversibility aligning with the principles of reversible logic. The controlled-NOT (CNOT) operation performed by Peres gates was examined through simulation, ensuring that the NOT operation is selectively applied based on the control bits. The controlled nature of the XOR operations ensures that information is not lost during the addition process. The overall functionality verification provides confidence in the correct operation of the designed RCA and multipliers using Peres gates. The subsequent sections delve into the performance analysis, including considerations of speed, quantum cost, and comparative assessments with traditional non-reversible designs. The 4-bit ripple carry adder shown in Figure 20 is constructed with four full adders where the carry is propagated through all the four full adders connected in series. The 4-bit carry look-ahead shown in Figure 21 consists of four full adders along with the carry propagator and carry generator circuit. The 4-bit carry-save adder shown in Figure 22 is constructed from eight full adders. Both carry look-ahead adders and carry-save adders have more quantum cost and occupy a larger area than ripple carry adders. The 4-bit carry skip or carry bypass adder shown in Figure 23 uses four full adders, one 4-bit AND gate, and a multiplexer, and the quantum cost for this is less than the carry look-ahead adder and carry-save adder.

From Figure 33, referring to 1-bit, 4-bit, 8-bit, and 16-bit adder configurations verified using RCViewer+ and ModeSim- Intel FPGA starter Edition 10.5b, we can easily develop the following expressions as given below.

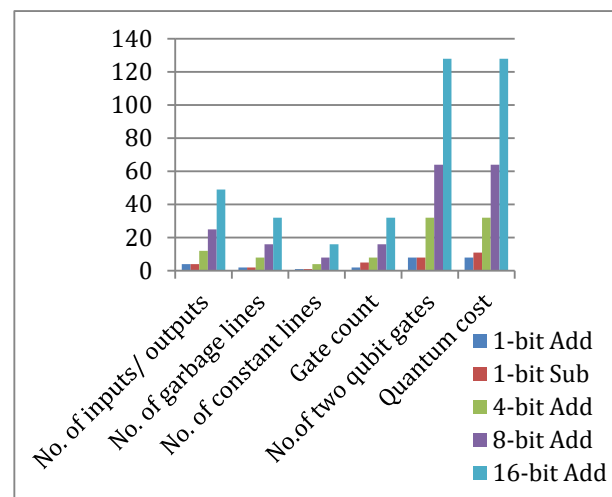


Figure 33. Performance analysis of different adder configurations using Peres gates.

- The number of inputs/outputs (n) is calculated as
- $N = n \cdot 3 + 1$
- where n-number of bits for the adder (in our case 1, 4, 8, 16, ...)
- The number of garbage lines (g) can be calculated as $g = 2 \cdot n$
- The number of constant lines $c = n$
- The number of gates (Gate count) required for constructing an adder is calculated as $GC = 2 \cdot n$
- The number of two-qubit gates (TQ) is given by $TQ = 8 \cdot n$

- And finally, the quantum cost (QC) is calculated by $QC = 8 \cdot n$

Hence, by using the above equations, we can easily verify the following further adder configurations, as shown in Figure 34.

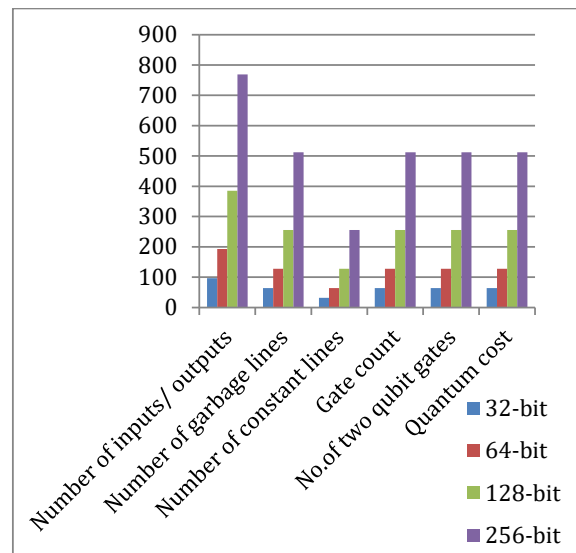


Figure 34. Further performance analysis of different adder configurations using Peres gates.

A comparative analysis was conducted between the proposed reversible design and other reversible designs available in the literature. This analysis considered factors such as gate count, energy consumption, and overall circuit complexity.

The goal was to highlight the potential advantages of the reversible approach, especially in terms of information preservation and resource utilization. Quantum speedup potential, which is a measure of how much faster a quantum algorithm could perform a specific task compared to its classical counterpart, was explored. The inherent reversibility of the designed circuit using Peres gates positions it favorably for potential quantum speedup in certain algorithms, contributing to the broader understanding of reversible logic in quantum computation. After comparing all the reversible adders using different techniques, we can say that the ripple carry adder is best suited due to the low metric values, as mentioned in Figure 35. The smaller number of inputs, constant and garbage lines, and quantum cost requirement to design an adder circuit are advantageous to using a ripple carry adder.

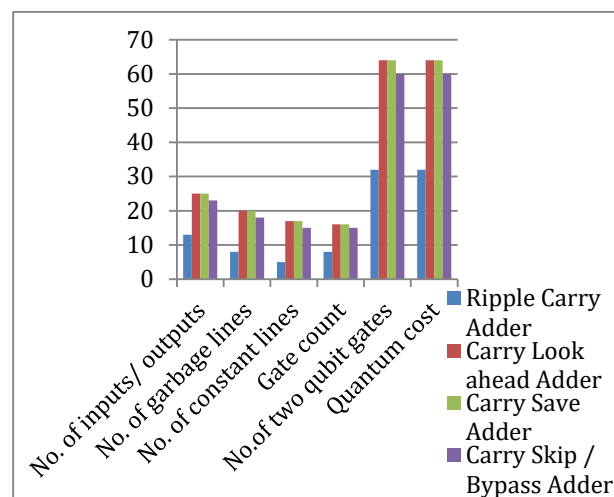


Figure 35. Analysis of Different Adder Configurations using Peres Gates.

In Table 2, various reversible logic gates used are Feynman Gate (FG), Toffoli Gate (TG), Fredkin Gate (FRG), Peres Gate (PG), New Gate (NG), TSG Gate (TSG), MKG Gate (MKG), HNG Gate (HNG), and PFAG Gate (PFAG) for comparison on the number of gates used for the construction of different reversible multiplier configurations. The first three multiplier circuits are designed as a part of research in this research article using only Toffoli and Peres gates. The other comparative circuits used other reversible gates available in the literature, as shown in the table. In Figure 36, the comparison of different multiplier circuits is made on the basis of quantum cost. The 4-bit reversible multiplier proposed in this article has a comparable quantum cost as compared to other multiplier circuits. The reduced quantum cost of 256 is the main outcome of this research as compared to a quantum cost for the same circuit design in the literature.

Table 2. Use of gates for different Multiplier configurations.

Multiplier Circuit	No. of Reversible Gates			
	Peres Gates	Toffoli Gates	Feynman Gates	Other Gates
Proposed design 1-bit	1	-	-	-
Proposed design 2-bit	2	4	-	-
Proposed design 4-bit multiplier using RCA	15	16	-	-
Proposed design 4-bit multiplier using CSA	16	16	-	-
Multiplier design in [26]	20	-	-	8 PFAG
Multiplier design I in [27]	11	9	-	8 PFAG-
Multiplier design II in [27]	11	9	-	8 HNG
Multiplier design in [28]	20	-	16	8 PFAG-
Multiplier design in [29]	16	-	16	12 MKG
Multiplier design in [30]	-	-	16	16 FRG + 13 TSG
Multiplier design in [31]	-	12	16	16 FRG + 12 NG

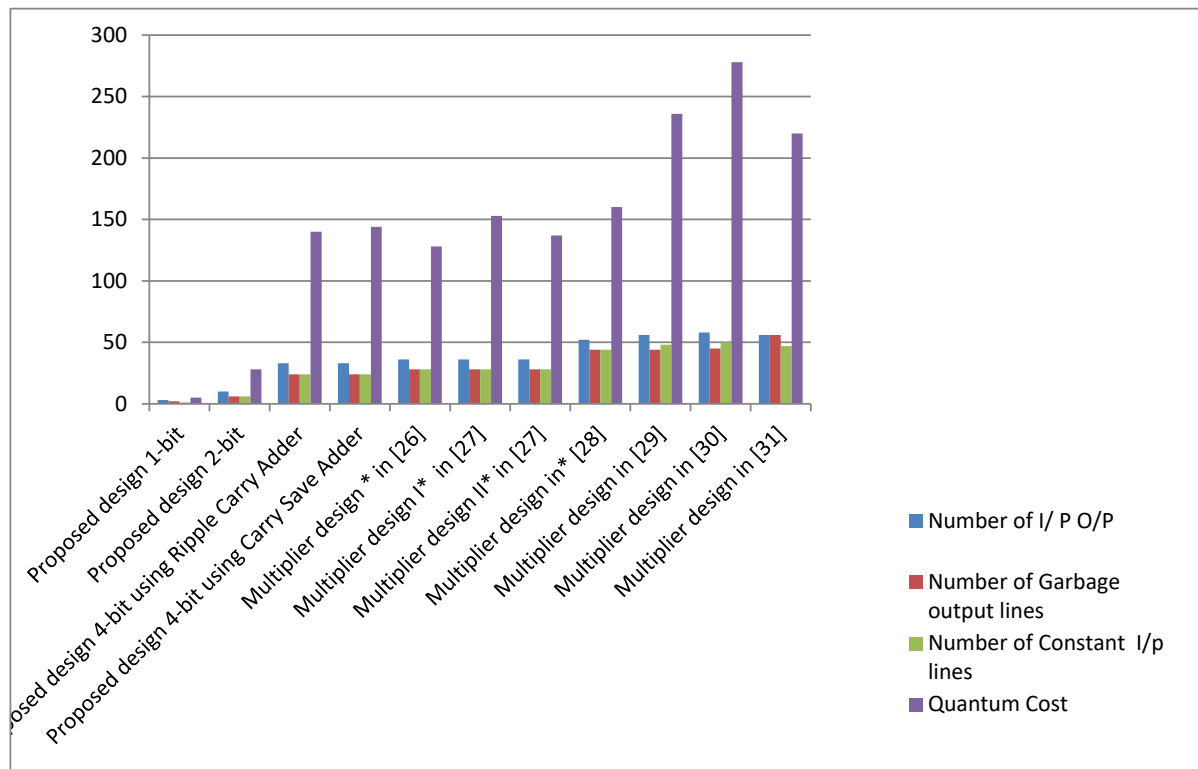


Figure 36. Performance analysis of different multiplier configurations.

Where * indicates the partial product output and not the final product of two 4-bit numbers in all the references, and hence, the quantum cost of these [26,27] circuits will be finalized after adding further stages in these circuits, and hence the quantum cost for the final product of the 4-bit multiplier using the ripple carry adder and carry-save adder is definitely greater than 140 and 144, which is the landmark for the proposed multiplier circuit. A 10% to 48% reduction in the quantum cost is observed, referring to the above table parameters in the proposed designs.

The analysis also considered potential optimization opportunities for the design. This includes exploring advanced reversible gates, circuit restructuring, or algorithmic enhancements to further improve performance metrics and address any identified limitations. The comprehensive evaluation of performance metrics provides valuable insights into the strengths and potential areas of improvement for the RCA design using Peres gates. The outcomes of these analyses contribute to the broader understanding of reversible logic circuits and their applications in emerging computational paradigms. In Figure 37, Levels 1, 2, and 3 on the Y-axis represent low, moderate, and high levels, respectively. From this figure, we can conclude that the Vedic multiplier is the best reversible multiplier with low gate count, quantum cost, gate level delay, hardware complexity, scalability, area, and number of Ancilla bits with high garbage outputs, which can be neglected as the reversibility efficiency provided by this circuit is high. The ripple carry adder constructed using reversible logic has all metrics considered to be at a moderate level, but it provides higher gate level delays, which can be neglected as the hardware complexity and area in terms of gates is also less. After examining all the adders and multiplier circuits using Toffoli and Peres gates, a comparison table for designing various adders and multipliers using reversible logic is prepared and presented in terms of a chart, as shown in Figure 37. The key parameters considered in the design of reversible circuits include all the metrics used to measure the performance of these circuits.

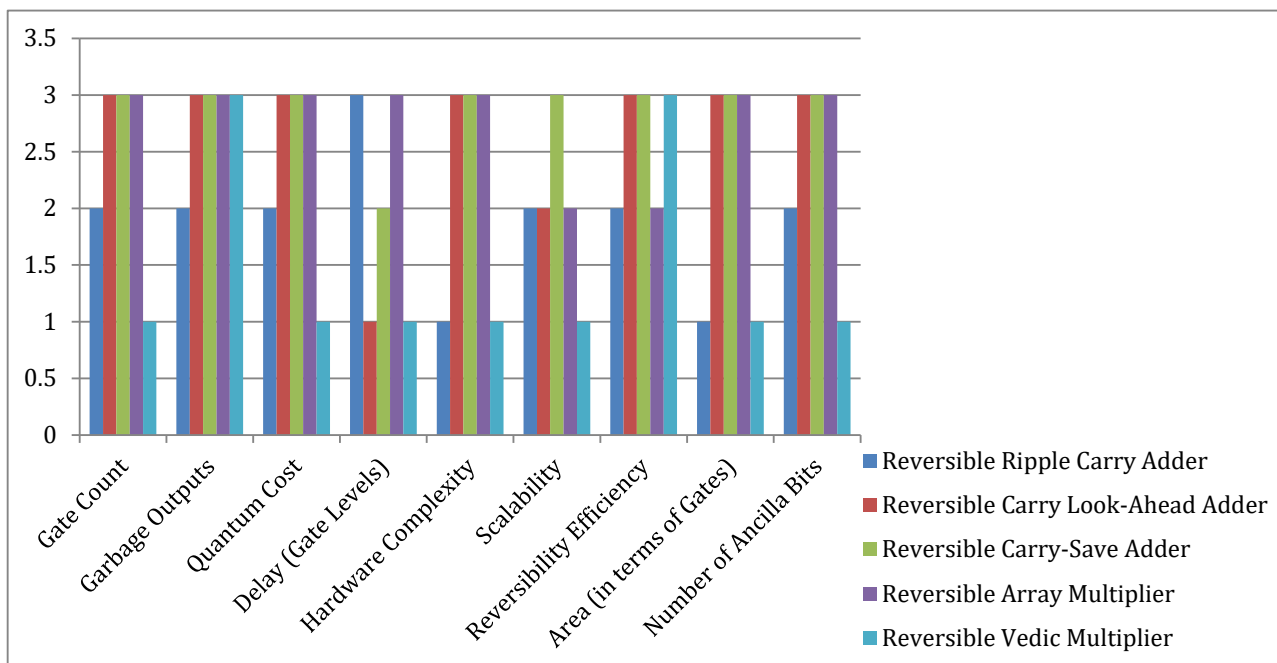


Figure 37. Comparative analysis of different adders and multiplier configurations.

When comparing the power needs of circuits based on Peres gates to those using other reversible logic gates, several factors must be considered, including gate design, energy consumption per operation, and overall circuit efficiency. Peres gates help to minimize the number of gates needed for certain operations, which lowers the switching activity and dynamic power consumption in circuits. Since fewer gates often result in fewer transitions,

this reduces overall energy usage. Hence, in quantum computing, implementing Peres gates sometimes requires fewer resources than traditional gates, potentially allowing for lower power requirements when executing some specific algorithms. Generally, the power consumption of reversible gates depends on the switching activity. The gates that require multiple transitions, like Toffoli gates, consume more power due to increased switching activity compared to a circuit optimized with Peres gates. Also, the circuits built using other reversible gates require overall more gates to achieve the same functionality, potentially leading to higher power needs. More gates mean more transitions and, therefore, greater dynamic power consumption, leading to better-optimized circuits. Peres gate-based circuits often have the potential to lower power consumption due to fewer transitions and gate count, but the specific performance depends on the design and implementation details of the overall circuit.

As a matter of fact, the reversible ripple carry adder is suitable for low-power applications but has a higher delay due to the sequential carry propagation. The reversible carry look-ahead adder is faster than ripple carry due to parallel carry calculation but requires more gates and garbage outputs. The reversible carry-save adder is used in multi-operand addition scenarios, like in multipliers, but with higher complexity. The reversible multiplier is a simple design but has a high gate count and quantum cost. The reversible adder and multiplier design using Vedic mathematics are optimized for high-speed operations but at the cost of significantly increased hardware complexity and power consumption.

Figure 38 shows the performance comparison between the different metrics used for different reversible logic gates used in this. The parameter levels are shown with bars of heights 1, 2, and 3 representing low, moderate, and high, respectively, for respective reversible logic gates. Fewer gates are required due to the Peres gate's dual function of AND and XOR, which further minimizes the quantum cost and also fewer garbage outputs than Toffoli and Fredkin-based designs. From this chart, it is quite clear that the use of the Peres gate for the design of adders and multipliers outperforms the other reversible gates.

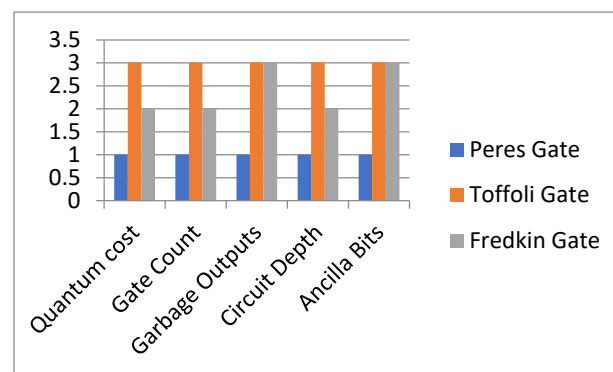


Figure 38. Performance Comparison Summary.

7. Future Considerations

While the current simulations provide promising results, further optimizations and exploration of advanced reversible logic gates could enhance the overall performance of the design. Additionally, the integration of the proposed design into larger circuits and quantum algorithms warrants investigation for its scalability and applicability. There might be different opportunities for optimizing the circuit design, including the investigation of advanced reversible gates and alternative architectures that may further enhance the efficiency and speed of the RCA. The integration of the designed circuit into quantum algorithms and assessing its compatibility and efficiency in specific quantum computing applications can be investigated. Explore how the reversibility of the circuit can contribute to quantum speedup in relevant algorithms. The scalability of the proposed design can be assessed by exploring its integration into larger circuits and arithmetic units.

These future considerations aim to extend the impact of the current research, fostering a deeper understanding of reversible logic circuits and their potential applications in the rapidly evolving fields of quantum and reversible computing.

8. Conclusions

The Peres gate-based designs for both adders and multipliers tend to outperform designs based on Toffoli and Fredkin gates in terms of quantum cost, garbage outputs, gate count, and circuit depth. Peres gate-based design is particularly useful in reversible computing, especially when optimizing for energy efficiency and minimizing resource overhead in quantum and low-power applications. The simulation and analysis affirm the functionality and potential advantages of the adders and multipliers designed using Peres gates and are efficient and information-preserving circuits. The reduction in the quantum cost of about 10% to 48% for 4-bit multipliers with ripple carry adder and carry-save adder using Peres gates comparing the literature circuits is the main outcome of this research.

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